Noninfectious Pulmonary Complications

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52.1 Introduction

Lung injury occurs frequently following HCT and significantly contributes to morbidity and mortality in the immediate posttransplant period and in the months and years that follow. In aggregate, pulmonary dysfunction can be observed in 25–55% of recipients (Cooke and Yanik 2016; Haider et al. 2020). In this context, an NIH workshop was recently convened to specifically identify clinical challenges and scientific knowledge gaps regarding pulmonary dysfunction in pediatric HCT recipient (Tamburro et al. 2021).

Historically, approximately half of all pulmonary complications seen after HCT were secondary to infection, but the judicious use of broad-spectrum antimicrobial agents has tipped the balance toward noninfectious causes.

Noninfectious lung injury following HCT may be mediated by immune or nonimmune mechanisms and accounts for up to 50% of noninfectious mortality after allo-HCT.

These complications have been classified by the American Thoracic Society according to the tissue primarily injured and its etiology (Panoskaltsis-Mortari et al. 2011) (Table 52.1).

Table 52.1 Noninfectious pulmonary complications after HCT[a]

Localization	Entity
Pulmonary parenchyma	– Acute interstitial pneumonitis[b] – Acute respiratory distress syndrome (ARDS)[b] – BCNU pneumonitis – Radiation pneumonitis – Delayed pulmonary toxicity syndrome[b] – Post-HCT lymphoproliferative disease (see Chap. 45) – Eosinophilic pneumonia – Pulmonary alveolar proteinosis
Vascular endothelium	– Peri-engraftment respiratory distress syndrome (PERDS)[b] – Capillary leak syndrome (CLS)[b] (see Chap. 42) – Diffuse alveolar hemorrhage (DAH)[b] – Pulmonary VOD – Transfusion-associated acute lung injury – Pulmonary cytolytic syndrome – Pulmonary arterial hypertension – Pulmonary thromboembolism

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Table 52.1 (continued)

Localization	Entity
Airway epithelium	– Interstitial lung disease including cryptogenic organizing pneumonia (COP)[b, c] – Bronchiolitis obliterans syndrome (BOS)

[a] Importantly, this classification does not include the most frequent lung complication after HCT, i.e., pulmonary edema secondary to fluid overload
[b] All these complications could be categorized as IPS if occurring early after HCT
[c] Formerly called bronchiolitis obliterans organizing pneumonia (BOOP)

52.2 Approach to Diagnostic Workup for Pulmonary Complications After HCT

Ideally, any respiratory/pulmonary complication observed after HCT whether early or late onset should be evaluated following predetermined, institutional, standard of practice guidelines (Lucena et al. 2014). Workup requires attention to pulmonary and non-pulmonary causes. Respiratory distress may progress rapidly once identified; hence, a timely workup including assessment of pulmonary and cardiac function, imaging, and procurement of samples to rule out infection is critical for optimizing outcomes (Tamburro et al. 2021; Cooke and Yanik 2016):

1. *Assessment of oxygen saturation/oxygen requirement*: Pulse oximetry or arterial blood gas analysis if indicated. For late-onset noninfectious pulmonary complications (LONIPC), pulse oximetry following a hall walk is suggested.
2. *Imaging*: Chest radiographs (particularly in children) or CT scanning to assess for the presence of lobar, multi-lobar, or diffuse pulmonary infiltrates. For early-onset lung injury, a low-dose CT can suffice. For LONIPC, a CT with inspiratory cuts is mandatory, and expiratory views to assess for air trapping should be discussed. Echocardiography can be considered to identify left-heart dysfunction or pulmonary hypertension.
3. *Pulmonary function testing* (PFTs): PFTs are rarely indicated in the context of acute lung complications but can be considered. For LONIPC, PFTs including spirometry (without and with albuterol), lung volumes, and DLCO are recommended to assess for signs for either restrictive or obstructive pulmonary dysfunction and to assess severity (Jagasia et al. 2015; Wolff et al. 2021). Assessing PFTs in pediatric patients remains a challenge particularly in young children (Tamburro et al. 2021).
4. *Infectious disease testing*: Blood samples for bacterial culture and detection of viral replication; sputum culture; nasopharyngeal swabs for rapid viral panels including SARS-CoV-2, respiratory syncytial virus (RSV), parainfluenza virus (PIV), adenovirus (ADV), rhinoenteroviruses, influenza A and B (in season), metapneumovirus, and others; urinary antigen tests; blood samples for galactomannan and beta-D-glucan assays.
5. *Fiber-optic bronchoscopy (FOB) with bronchoalveolar lavage (BAL)* (strongly encouraged): Complete in collaboration with specialists in pulmonary medicine and infectious diseases. PCR for *Pneumocystis jirovecii* (PJ), *Legionella*, and *Mycoplasma*. Cultures and stains for bacteria, fungi, and AFB along with galactomannan assay. If BAL cannot be completed, consider approaches to induce and collect sputum for the assays noted above:
   - If positive, treat accordingly.
   - If negative, consider empiric treatment and consideration for idiopathic pneumonia syndrome (IPS) and other etiologies (see below).
6. *Tissue biopsy*: While not encouraged, in some selected cases, e.g., later-onset cases where the diagnosis is not established by noninvasive tests or those poorly responsive to antimicrobial or anti-inflammatory therapy (see below), a tissue biopsy could be considered using either transbronchial (less favored) or minimally invasive, video-assisted thoracoscopic surgery (VATS) (Dieffenbach et al. 2019).

52.2.1 Results Reported Using This Methodology (Seo et al. 2015; Lucena et al. 2014; Shannon et al. 2010; Yanik et al. 2008a, b)

FOB/BAL permits an etiological diagnosis in up to 78% of cases.

In suspected IPS, a BAL study may detect a pathogen in ~50% of cases.

For pathogen detection, early FOB (<5 days) offers better yield than late FOB; hence, completing FOB as soon as possible is preferred.

The risk of complications with FOB is <5%.

Note: Identifying an organism in pulmonary secretions does not always imply causality, and infectious and immune-mediated inflammation can occur concurrently.

52.3 Acute Noninfectious Pulmonary Complications After HCT

52.3.1 Pulmonary Edema due to Fluid Overload

Despite not being included in most classifications of pulmonary complications after HCT, pulmonary edema (PE) as a consequence of a fluid overload (FO) is extremely frequent (Rondón et al. 2017):

Incidence	FO may be observed in up to 60% of patients early after HCT. The exact incidence of PE is not established although it could be higher than 20%.
Symptoms and signs	– Weight gain, shortness of breath, nonproductive cough, moderate hypoxemia. – Crackles and rales in both lung bases. – Chest radiology with diffuse alveolar/interstitial infiltrates.
Diagnosis	PE should be suspected in the context of weight gain, an increased cardiothoracic index on imaging, and crackles/rales. Though rarely necessary, the diagnosis can be confirmed by pulmonary pressure measurements.
Differential diagnosis	– Heart failure (prior anthracycline toxicity or conditioning with CY). – Endothelial syndromes: SOS, CLS, ES (see Chaps. 42 and 49). – Respiratory tract infections: • Idiopathic pneumonia syndrome (IPS) – Posttransfusion reactions (TRALI).
Treatment	Goal: Re-establish fluid homeostasis/euvolemia via fluid restriction and diuretics.

52.3.2 Idiopathic Pneumonia Syndrome

52.3.2.1 Definition

It is a widespread alveolar injury in the absence of active lower respiratory tract infection, cardiac or renal dysfunction, and iatrogenic fluid overload (Clark et al. 1993; Panoskaltsis-Mortari et al. 2011).

52.3.2.2 Clinical Manifestations

Signs and symptoms of IPS classically occur within 120 days (but as early as 3 weeks and as late as 180 days) after HCT and include fever, nonproductive cough, dyspnea, tachypnea, hypoxemia, rales, and diffuse alveolar or interstitial infiltrates on X-rays or CT scans.

52.3.2.3 Diagnosis

All of the following must be present for the diagnosis of IPS:

1. *Evidence of widespread alveolar injury:*
(a) Multi-lobar infiltrates on chest radiographs or CT (b) Symptoms and signs of pneumonia (cough, dyspnea, tachypnea, crackles/rales) (c) Evidence of abnormal pulmonary physiology Increased alveolar to arterial oxygen difference; need for supplemental O_2 therapy New or increased restrictive PFT abnormality
2. *Absence of active lower respiratory tract infection based upon:*
(a) BAL negative for significant bacterial pathogens including acid-fast bacilli, *Nocardia*, and *Legionella* species (b) BAL negative for pathogenic nonbacterial microorganisms Routine culture for bacteria, fungi, and AFB and shell vial culture for CMV PCR for respiratory viruses and CMV Cytology for CMV inclusions, fungi, and *Pneumocystis jirovecii* (c) Other organisms/tests to also consider: PCR for human metapneumovirus, rhinovirus, coronavirus, and HHV6 PCR for *Chlamydia*, *Mycoplasma*, and *Aspergillus* spp. Serum and BAL fluid GM for *Aspergillus* species, serum beta-D-glucan
3. *Absence of cardiac dysfunction, acute renal failure, or iatrogenic fluid overload as etiology for pulmonary dysfunction*

52.3.2.4 Pathogenesis, Incidence, Presentation, and Risk Factors

Pathogenesis	– The pathophysiology of IPS is complex. Data generated using experimental models support IPS as a process in which the lung is susceptible to cellular and soluble inflammatory effectors. These distinct but related pathways of inflammation culminate in the recruitment of immune cells to the lung leading to tissue damage and dysfunction (Cooke and Yanik 2016). – Recent work has also underscored a role for the pulmonary microbiome and metatranscriptome to the development of lung inflammation after HCT (Zinter et al. 2021).
Biomarker incidence	– sTNFR1, IL-6, MCP-1, Ang2, sCD14, and IL-8 are elevated in patients with IPS (Yanik et al. 2008a, b, 2015). – sTNRF1, IL-6, and ST-2 were most predictive in diagnosing IPS even before clinical signs and symptoms were present (Seo et al. 2017). – In addition, a biomarker panel including ST2, IL-6, and sTNFR1 could, as early as day 7, predict respiratory failure and associated mortality after HCT (Rowan et al. 2022). – The strict definition required to establish a diagnosis and the increased use of RIC have significantly reduced the incidence of IPS from that observed in the past (~25%). – This reduction runs in parallel of the improvement in the diagnostic methodologies to detect infectious pathogens (Seo et al. 2015; Zinter et al. 2019). – Currently, the incidence of IPS is <10% of allo-HCT (7% after MAC with ≥12 Gy TBI; 2% after RIC in adults) (Wenger et al. 2020; Sano et al. 2014).
Timing	– Generally, within first 120 days after BMT but can be observed between days 21 and 180. – Late IPS (up to180 days post-HCT) can be observed but is rare (Thompson et al. 2017).
Risk factors (from Cooke and Yanik 2016)	– Older age/Karnofsky index <90/ higher interval diagnosis HCT. MAC or TBI (≥12 Gy)/HLA disparity/ GVHD prophylaxis with MTX. Acute GVHD/previous viral infection/ other malignancies than leukemia.

52.3.2.5 Treatment and Prognosis

Supportive measures	Supplemental O_2 therapy Noninvasive [high-flow nasal O_2, CPAP] or, when needed, invasive (mechanical) ventilatory support: – Empiric broad-spectrum antimicrobials – Strict control of fluid balance/ hemofiltration

Specific treatment	Lung injury in IPS can occur through two pathways, the TNF-alfa/LPS-dependent and IL6/IL17-dependent (Tamburro et al. 2021; Cooke and Yanik 2016; Varelias et al. 2015) informing current treatment options: • *Methyl-PDN* ≤ 2 mg/kg/day; if not clear response, consider as soon as possible. • Anti-TNFα: *etanercept* 0.4 mg/kg twice weekly (maximum of 8 doses) + systemic steroids (2 mg/kg/day). In a phase II trial in children, the CR rate was 71%, and 1 year survival was 63% (Yanik et al. 2015). This combination has also been shown to be effective in exceptional cases of late IPS with a 42% of CR and a 2-y survival of 62% among responders (Thompson et al. 2017). The randomized study of etanercept + steroids vs. steroids + placebo was terminated prematurely due to slow accrual. In the limited number of patients examined, there were no differences in response rates (≈60%) at day +28. These results do not necessarily imply that this agent is not effective; lack of evidence in this under-accrued trial does not equate to lack of effectiveness (Yanik et al. 2014). • Other *investigational agents* such as: – MoAb anti-IL6: tocilizumab (experimental IPS; Varelias et al. 2015). Note, IL-6 levels were found to be highest in patients not responsive to TNF blockade (Varelias et al. 2015) suggesting that strategies of combinatorial cytokine blockade may warrant future study (Tamburro et al. 2021). – MoAb anti-IL17: brodalumab (experimental IPS; (Varelias et al. 2015). – Strategies that protect/improve pulmonary vascular endothelial integrity (Klein et al. 2023).
Evolution	Despite the diagnosis and therapeutic advances, the mortality from IPS in adults remains high (Wenger et al. 2020).

52.3.3 Diffuse Alveolar Hemorrhage (DAH)

Diffuse alveolar hemorrhage (DAH) is a relevant cause of acute respiratory failure that occurs in 2–14% of recipients (Fan et al. 2020; Zhang et al. 2021). DAH is likely a consequence of damage to the alveolar capillary basement membrane (see Chap. 42). DAH can be noninfectious or infectious in etiology (Majhail et al. 2006).

52.3.3.1 Clinical Aspects of DAH

Clinical manifestations	Usually observed within the first month after HCT (a median of 23 days), often during the pre-engraftment phase; however, later onset is encountered in up to 42% of cases The clinical manifestations are those of all IPS. Frank hemoptysis is rare.
Diagnosis	Based on BAL: the progressive bloodier return of BAL fluid aliquots, in at least three segmental bronchi, indicating the presence of blood in the alveoli (or 20% hemosiderin-laden macrophage, although their absence does not exclude the diagnosis as it can take 72 h to appear).
Risk factors	– Higher incidence after TBI and high-dose CY.
Differential diagnosis with	– Classic IPS: very difficult, only by means of BAL. IPS usually appears after engraftment, predominates in allo-HCT, and is less responsive to steroids, *Note*: Noninfectious DAH falls under the "diagnostic umbrella" of IPS (Panoskaltsis-Mortari et al. 2011). – PERDS: almost impossible except for LBA progressively bloodier. – DAH associated with infection: impossible without detection of the pathogen (Majhail et al. 2006).

52.3.3.2 Treatment and Prognosis of DAH

Treatment	– Traditionally with high doses of methyl-PDN (250–500 mg q6h × 5 days followed by taper), but the overall response to this treatment is disappointing even when combined with aminocaproic acid (ACA) (Rathi et al. 2015). Of note, two-thirds of cases were noninfectious in origin and could be classified as IPS, but TNFa inhibition was not used in this study. – When combined with high-dose PDN, inhaled, recombinant, factor VIIa has shown benefit in some (Park and Kim 2015) but not all (Elinoff et al. 2014) studies.

Prognosis	– Poor: Overall mortality as high as 85% by day 100 (Rathi et al. 2015). – Less than 15% of patients die as a direct consequence of DAH, but the frequent evolution to MOF increases mortality to >60% (30% in auto- and 70% in allo-HCT). – DAH that appears after auto-HCT has a better prognosis (Majhail et al. 2006).

52.4 Late-Onset Noninfectious Pulmonary Complications (LONIPC)

LONIPCs, when defined as occurring beyond 100 days posttransplant, occur in up to 20% of allo-HCT recipients, including a variety of clinical entities, with bronchiolitis obliterans syndrome (BOS) and diffuse interstitial lung diseases (ILD) being the most frequent (Bergeron et al. 2018). Before making the diagnosis of LONIPC, a respiratory infection must be ruled out. However, a respiratory infection (i.e., particularly viral) is often the trigger for LONIPC. Although BOS is currently recognized as the only manifestation of chronic GVHD based on epidemiological data, the question arises whether ILD may be another lung manifestation of cGVHD (Archer et al. 2023).

52.4.1 Bronchiolitis Obliterans Syndrome (BOS)

Pathogenesis, timing, incidence, clinical manifestations, diagnosis, and radiology of BOS are shown in Table 52.2.

Table 52.2 Main clinical characteristics of BOS

Pathogenesis Risk factors	The same as cGVHD but specifically involving the lung (Cooke et al. 2017). Extrathoracic cGVHD, respiratory viral infections, lung function decline (Bergeron et al. 2018; Sheshadri et al. 2019; Jamani et al. 2020; Abedin et al. 2015).
Timing and incidence	Onset after transplant: 6–24 months. Incidence: 3–11%.
Clinical manifestations	– Breathlessness, nonproductive cough, and wheezing. – May be asymptomatic requiring routine PFT follow-up for an early detection (pretransplant baseline, then every 3 months in the first year after HCT and in patients who develop extrathoracic GVHD; close spirometry monitoring after a respiratory viral infection (Kitko et al. 2021)). In >90% of the BOS, there is more or less severe chronic GVHD in other locations.
Diagnosis	Chronic GVHD in other locations[a] + PFT impairment (NIH consensus) (Jagasia et al. 2015; Wolff et al. 2021). – New-onset obstructive ventilatory defect with air trapping (FEV1/VC ratio < 0.7 or the fifth percentile of predicted; FEV1 < 75% of predicted with ≥10% decline from baseline; residual volume > 120% of predicted) with no response to bronchodilators. • *Definitive* (not required): histologic confirmation by VATS (BO).
Lung CT scan prognosis	Required at BOS diagnosis to rule out other diagnoses (lung parenchymal abnormalities should suggest either an alternative or an additional diagnosis). Pattern of constrictive bronchiolitis with attenuation in mosaic, bronchiectasis, and bronchial wall thickening and air trapping by expiratory CT[b] Bronchial abnormalities at 3 months posttransplant were found to predict BOS (Bergeron et al. 2018). Overall survival: 65–79% at 2 years and 74% at 5 years in the most recent studies (Arora et al. 2016; Archer et al. 2023). Worse prognosis if early onset after HCT (Bergeron et al. 2013). Morbidity: respiratory disability, respiratory exacerbations.

DLCO transfer capacity of CO, *FEV1* forced expiratory volume in the first second, *FVC* forced vital capacity, *VATS* video-assisted thoracoscopic surgery, *CT* computed tomography

[a] If the lung is the only organ with cGVHD, a biopsy is needed to confirm the diagnosis

[b] More than 30% of patients have air trapping on pretransplant lung CT scan limiting the value of expiratory chest CT in the diagnosis of posttransplant BOS (Bergeron et al. 2018). BOS diagnosis mainly relies on PFTs

The latest NIH consensus does not require lung biopsy; the diagnosis of chronic pulmonary GVHD is made on the basis of lung function data alone. Thus, we speak of BOS rather than BO (i.e., constrictive bronchiolitis), which requires histological analysis (Jagasia et al. 2015). BOS phenotypes not included in the current NIH diagnostic criteria do exist. On the one hand, BOS reflects various pathologies of the small airways (Meignin et al. 2018; Holbro et al. 2013), and on the other, PFTs of BOS may show a preserved FEV1/VC ratio (Bergeron et al. 2013; Wolff et al. 2021; Uhlving et al. 2015).

52.4.1.1 Treatment of BOS

Few treatments have been prospectively and specifically evaluated for BOS, including those for chronic GVHD. Low levels of evidence prevent recommendation of any specific treatment algorithm. It is essential to weigh the benefit-risk balance before introducing any treatment, particularly corticosteroids, which are known to be of low efficacy for BOS.

The natural history of BOS is poorly understood (Cooke et al. 2017). FEV1 most often declines abruptly before stabilizing at a more or less severe value (Cheng et al. 2016; Bergeron et al. 2013, 2018). Secondary decline in lung function over time is often associated with infectious exacerbations. In this context, the endpoint for assessing the efficacy of any treatment on BOS is debated: stabilization or improvement of pulmonary function.

Only inhaled budesonide/formoterol has shown improvement of FEV1 for new-onset moderate to severe BOS in a double-blind placebo-controlled randomized trial (Bergeron et al. 2015). The FAM strategy (fluticasone, azithromycin, montelukast) that is commonly used for the treatment of BOS was evaluated in an open-label single-arm trial. In this study, FAM was associated with a stabilization of FEV1 (Williams et al. 2016). Azithromycin administration in allo-HCT recipients should be cautioned against a potential increased risk of cancer, particularly when given early after transplantation (Bergeron et al. 2017; Cheng et al. 2020; Vallet et al. 2022).

Current diagnostic criteria for BOS probably only allow detection of a late stage of the disease. Numerous studies are looking for early biomarkers of BOS (respiratory function, and/or radiology, and/or biology) in the hope of greater treatment efficacy at an earlier stage.

In patients with uncontrolled end-stage BOS, who have no or non-active extrathoracic GVHD and very low risk of hematological relapse, lung transplantation should be considered (Greer et al. 2018).

Respiratory rehabilitation should be offered to dyspneic patients to improve their quality of life. Other support measures include treatment of gastroesophageal reflux, anti-infectious prophylaxis, vaccination against respiratory viral infections, and IVIg in case of both infections and hypogammaglobulinemia.

52.4.2 Interstitial Lung Diseases (ILD)

Three-year cumulative incidence of ILD was found to be 5% (Bergeron et al. 2018). Unlike BOS, the diagnosis of ILD relies on imaging showing the presence of lung parenchymal opacities in the absence of infection. Based on lung CT scan, ILD could be classified into three patterns: organizing pneumonia (OP, 37%), pleuro-parenchymal fibroelastosis (PPFE, 13%), and undetermined (50%) (Archer et al. 2023; Bondeelle et al. 2020). The "undetermined" CT scan pattern corresponds to different clinico-histological diagnoses, including nonspecific interstitial pneumonia (Meignin et al. 2018). The clinical presentation may be acute, subacute, or chronic; in a quarter of cases, the diagnosis is made in intensive care (Archer et al. 2023).

Unlike BOS, which is exceptionally diagnosed more than 2 years after transplantation, ILD can occur anytime. Of note, PPFE occurs several years after transplantation (Bondeelle et al. 2020). Most patients present with dyspnea.

Specific risk factors for ILD are prior thoracic irradiation and the absence of immunosuppressive treatment at the time of ILD occurrence (Archer et al. 2023).

In the context of ILD, complete PFTs (including lung volumes and DLCO) are useful for assessing the severity of lung dysfunction and monitoring treatment efficacy. PFTs also allow to diagnose combined ILD and BOS which occurs in 25% of cases of ILD (Archer et al. 2023).

A distinction should be made between inflammatory ILD (OP and undetermined), which are usually sensitive to corticosteroid therapy, and PPFE, which have low inflammation and do not respond to immunosuppressive treatments, including steroids (Table 52.3).

Table 52.3 Comparative characteristics between BOS and ILD

Median onset after HCT	BOS: 13 months ILD: 15 months[a]
Extrathoracic cGVHD	BOS: >90% ILD: >75%[b]
Thoracic CT scan[c]	BOS: bronchial abnormalities, air trapping, *no* alveolar and/or interstitial abnormalities ILD: alveolar and/or interstitial abnormalities
PFT pattern[c]	BOS: obstructive pattern – FEV1/FVC < 0.7 and TLC > 80% ILD: normal/restrictive pattern, TLC <80% and FEV1/FVC > 0.7; mixed ventilatory defect[d], TLC < 80% and FEV1/FVC < 0.7; DLCO reduced
Treatment	BOS: limited ILD: usual response to steroids[e], risk of relapse when steroids are tapered off
Prognosis[f]	BOS: 3-year OS 79%; 5-year OS 74% ILD: 3-year OS 81%; 5-year OS 71%

PFT pulmonary function testing, *PPFE* pleuroparenchymal fibroelastosis, *FEV1* forced expiratory volume in 1 s, *FVC* forced vital capacity, *TLC* total lung volume, *DLCO* diffusion capacity of the lungs for carbon monoxide, *OS* overall survival

[a] PPFE occurs several years after HCT

[b] Less than 50% in PPFE

[c] The diagnosis of BOS is based on PFT, whereas those of ILD relies on chest imaging

[d] When combined with BOS

[e] Except for PPFE where the treatment is limited

[f] Morbidity is significant including respiratory disabling and infections

Key Points

- Noninfectious lung injury occurs frequently following HCT and significantly contributes to morbidity and mortality in the immediate posttransplant period and in the months and years that follow. It can be observed in 25–55% of recipients.
- Noninfectious lung injury following HCT may be mediated by either immune or nonimmune mechanisms and could represent up to the 50% of noninfectious mortality after allo-HCT.
- Most relevant noninfectious early pulmonary complications are pulmonary edema by fluid overflow, IPS, and DAH, a vascular endothelial syndrome.
- The most relevant LONIPCs are BOS and ILD including OP.
- All of them have specific diagnostic criteria, management, treatment, and prognosis.

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